CARD 1 OF 8 ANNUAL TEST **QUESTION**

What test does NFPA 25 require on a fire protection **backflow preventer every year**?

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CARD 2 OF 8 HOSE STREAMS **QUESTION**

There are **hydrants downstream** of the backflow preventer. Does that change the forward flow test?

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CARD 3 OF 8 CAN'T REACH DEMAND **QUESTION**

The piping won't let you flow full system demand through the backflow preventer. **Now what?**

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CARD 4 OF 8 THE PUMP COVERS IT **QUESTION**

The annual **fire pump test** already pushes system flow through the backflow preventer. Do you still need a separate forward flow test?

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CARD 2 OF 8 HOSE STREAMS **ANSWER**

Yes. Add the hose stream demand.

If hydrants or inside hose stations are fed through the backflow preventer, it has to pass the sprinkler demand **plus** the hose demand. The test flow has to include both.

NFPA 25 (2020) · Ch. 13

On the job: a backflow preventer on a private fire main feeding yard hydrants. Pull the hydraulic calc and add the hose number before you set up the test.

CARD 1 OF 8 ANNUAL TEST **ANSWER**

A forward flow test at no less than the **system demand** flow rate.

The point: water has to get **through** the backflow preventer in a fire. The forward flow test proves the check valves open fully and pass the flow the sprinklers need.

NFPA 25 (2020) · Ch. 13

Trap: "The water company's backflow tester was out last month." That gauge test checks the assembly **holds back** water. It doesn't flow system demand, so it isn't the NFPA 25 forward flow test.

CARD 4 OF 8 THE PUMP COVERS IT **ANSWER**

No. If the annual pump test sends the system flow rate through the backflow device, that counts. A separate forward flow test isn't required.

Check first: the backflow preventer has to be **in the flow path** of the pump test, at system flow.

NFPA 25 (2020) · Ch. 13

Trap: a pump test header that tees off **upstream** of the backflow preventer doesn't flow through it, so it doesn't count.

CARD 3 OF 8 CAN'T REACH DEMAND **ANSWER**

Flow the most the connections allow. Test at the maximum flow rate possible.

Ways to get the flow (Annex):

- A **test header** or other connection downstream.
- A **bypass** around the FDC check valve, with a normally closed control valve.
- **Closed loop**, if a flowmeter or sight glass proves water is actually moving.

NFPA 25 (2020) · Ch. 13 · Annex A

Trap: "No test header, so no test." Flow what you can and record it.

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 WATER RATIONING **QUESTION**

Is there any situation where you can **skip the annual forward flow test** and do something else instead?

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CARD 6 OF 8 5-YEAR INTERNAL **QUESTION**

How often does a backflow preventer get an **internal inspection**, and what are you checking?

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CARD 7 OF 8 RPZ WEEKLY **QUESTION**

What do you look for on a **reduced-pressure (RPZ)** assembly every **week**?

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CARD 8 OF 8 AFTER THE WORK **QUESTION**

The backflow preventer was just tested or repaired. **Who checks it afterward**, and what are they looking for?

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CARD 6 OF 8 5-YEAR INTERNAL **ANSWER**

Every 5 years.

Checking: every part works correctly, moves freely and is in good condition.

Annex: if the annual maintenance already includes an internal inspection by a qualified person, that covers it.

NFPA 25 (2020) · Ch. 13 · Annex A

On the job: rebuilding the checks with a repair kit during maintenance? Write the internal inspection on the report. That's your 5-year.

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CARD 5 OF 8 WATER RATIONING **ANSWER**

Yes: water rationing lasting more than 1 year.

During a long shortage, an **internal inspection** that confirms the check valves will open fully can take the place of the annual forward flow test.

Otherwise, you flow it.

NFPA 25 (2020) · Ch. 13

Trap: "The city doesn't like us wasting water." Not the same as enforced rationing over a year.

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CARD 8 OF 8 AFTER THE WORK **ANSWER**

The property owner or their representative.

Checking: the system is back in service, and every **isolation valve** is **open** and **locked or electrically supervised**.

Maintenance itself: a **qualified** person does it, to the manufacturer's instructions and the AHJ's procedures.

NFPA 25 (2020) · Ch. 13

On the job: the tester closes both OS&Ys to test the checks. If one stays shut, the whole building is unprotected. That's what this check is for.

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CARD 7 OF 8 RPZ WEEKLY **ANSWER**

The relief port. It shouldn't be **discharging continuously**.

This applies to reduced-pressure assemblies and reduced-pressure **detector** assemblies.

Why: a relief port that keeps running is telling you a check valve isn't sealing.

NFPA 25 (2020) · Ch. 13

Trap: a **double check** doesn't have a relief port. The weekly relief-port look is for **RPZ** types.

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